SCT v2.2-F - Numerical and Formula Invariance Audit

Purpose

This audit checks that the v2.2-E centerless timeline-bundle reformulation changes public wording only and does not change formulas, tables, classifications, transfer results, or numerical claims from the v2.1-H public package.

Overall decision

PASS_NUMERICAL_INVARIANCE_AUDIT_WITH_QA_MANIFEST_NOTE

No numerical result changed. The only non-PASS item is a recorded v2.1-H QA-only manifest mismatch on qa/final_zip_test.txt.

Check matrix

Check	Status
package_manifest_v2_1_H	FAIL_QA_ONLY_KNOWN_PACKAGING_MISMATCH
package_manifest_v2_2_E_manual	PASS
v1_4_F_reference_vs_rerun	PASS_CORE_COUNTS_AND_TABLES
v1_6_E_reference_vs_rerun	PASS
v2_0_B_transfer_table_vs_rerun	PASS
v2_0_E_transfer_table_vs_numba_rerun	PASS
v2_2_E_no_numeric_replacement_payload	PASS

Key numerical results

Metric	Value
v1.4-F total reproduction	ALL REPRODUCIBILITY CHECKS PASS
v13B_rows	4320
v13B_ok_toy_total	681
v13C_rows	5040
v13C_coherent_toy	228
v13C_weak	2357
v13C_critical	1378
v13C_failed	1077
v1.6-E rows	140
v1.6-E coherent_toy	25
v1.6-E critical	28
v1.6-E failed	35
v1.6-E weak	52
v2.0-B alpha 0.1 classification	NON_GAUSSIAN_TRANSFER
v2.0-B alpha 0.1 max T diff	1.110e-16
v2.0-B alpha 0.2 classification	NON_GAUSSIAN_TRANSFER
v2.0-B alpha 0.2 max T diff	5.551e-17
v2.0-B alpha 0.5 classification	NON_GAUSSIAN_TRANSFER

Metric	Value
v2.0-B alpha 0.5 max T diff	9.714e-17
v2.0-E alpha 0.2 classification	NON_GAUSSIAN_TRANSFER
v2.0-E alpha 0.2 max T diff	2.670e-14
v2.0-E alpha 0.1 classification	NON_GAUSSIAN_TRANSFER

Formula invariance

Cyclic phase / unwrapped time: phi in S^1; tau in R; phi(tau)=tau mod 2pi, or normalized equivalent.

Entropy decomposition: S_abs(n,u)=S_min(n)+Delta S_n(u); S_rel(n,u)=Delta S_n(u).

Entropy floor: S_min(n+1) >= S_min(n).

Capacity diagnostic: In the flat-FLRW toy branch, normalized accessible capacity remains proportional to H^{-2}.

Entropy clock: alpha_eff(u)=Gamma_S(u)/H(u).

Transfer grammar: delta_(n+1)(k)=T_R(k) delta_n(k)+xi_n(k); P_(n+1)=|T_R|^2 P_n + P_xi under the stated assumptions.

v2.2 wording patch: The auxiliary sphere/common-point language is replaced by centerless timeline-bundle wording; this changes the diagram interpretation, not any operational equation.

Detailed rerun findings

v1.4-F reruns pass the shipped reproducibility check. Some published summary CSVs differ only by row order or by column naming in regenerated compact summaries; mapped counts and fractions match.

v1.6-E scan and classification counts match. The reason-count table differs only by row order before sorting.

v2.0-B and v2.0-E transfer tables reproduce to floating-point precision and preserve NON_GAUSSIAN_TRANSFER classifications.

Claim boundary

This audit is not a physical validation of SCT, not an observational test, not a restart derivation, and not a universal transfer-kernel derivation. It only confirms that v2.2-E does not alter the earlier numerical/formula content.